

Manikandan P

Full-Stack Engineer | Node.js · TypeScript · React · Event-Driven Systems

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SUMMARY

Full-stack engineer with close to 4 years building production systems on Node.js/TypeScript and React. Depth in event-driven backends (Kafka, BullMQ, MQTT, WebSockets), multi-tenant SaaS architecture, and AWS media infrastructure. Shipped fintech transaction platforms processing million-record reporting jobs and a multi-tenant ERP spanning three web portals and a mobile app. **Available to join immediately.**

SKILLS

Languages	TypeScript, JavaScript (ES2022), Node.js, Python, Bash
Backend	Fastify, Express, REST API design, Microservices, Event-driven architecture, Socket.IO, Worker threads, OpenAPI/Swagger, Zod, JWT, TOTP MFA
Data & Messaging	MongoDB (aggregation, indexing, cursors, 2dsphere geo), Mongoose, Redis, Apache Kafka, MQTT, BullMQ
Frontend	React 19, Next.js 15 (App Router), TanStack Query, Zustand, Redux Toolkit, Tailwind CSS, Radix UI
Mobile	React Native, Expo (SDK 54), Expo Router
Cloud & DevOps	AWS (S3, MediaConvert, IVS, ECR, SES, IAM), Docker (multi-stage), Kubernetes, Jenkins, GitHub Actions, Nginx
Testing	Playwright (E2E), Node test runner
Architecture	Multi-tenant SaaS, RBAC/IAM design, Finite state machines, Distributed job orchestration, Graceful degradation

EXPERIENCE

Gravitorix Technologies, Software Engineer

May 2026 - Present · Remote

Services company. Owned platform engineering across multiple client products.

- Architected and built a multi-tenant, white-labelled school ERP end to end: a Fastify/MongoDB backend (202 endpoints, 42 models), three Next.js 15 portals, and an Expo mobile app across 6 repositories, totalling roughly 70,000 lines of TypeScript.
- Eliminated cross-tenant data-leak risk by enforcing tenant scoping inside Mongoose middleware backed by AsyncLocalStorage request context, fail-closed by default with named, auditable escape hatches, replacing per-query developer discipline with a structural guarantee.
- Designed a dependency-closed IAM system (19 permission modules, 39 actions, 13 role templates) applying transitive grant-expansion and revoke-contraction identically in the admin UI and server-side on every write, plus a deploy-gating invariant checker that catches permission-graph drift as the catalog grows.
- Built a distributed driver-dispatch engine for a ride-hailing platform as a durable BullMQ state machine: geospatial matching via a compound 2dsphere index, 5-second offer timeouts, decline-driven re-matching with an exclusion accumulator, and dual attempt-count and wall-clock termination, surviving pod restarts where an in-process timer could not.
- Delivered an AI virtual try-on platform (Next.js 15, Fastify, Google Vertex AI): measured a first-pass MediaPipe pose-warp renderer at roughly 30% believable output, drove the pivot to a diffusion model, reached roughly 95% client-validated accuracy, then repurposed MediaPipe as a client-side pose-validity gate so unusable photos never reach the paid API.

Neokred Technologies, Software Development Engineer, Full-Stack

Nov 2023 - Apr 2026 · Bengaluru

Fintech and Banking-as-a-Service platform.

- Built the event-driven expense-processing pipeline for a B2B2C expense management product: ingested raw user messages onto Kafka, integrated an ML inference service across two topics, and persisted structured transactions powering category-level spending analytics, with response caching and failure handling across the asynchronous pipeline.
- Owned merchant transaction reporting in a UPI Switch platform, generating Excel reports of up to roughly 1,000,000 records without blocking the event loop, using BullMQ background jobs, Node.js worker threads, MongoDB cursors, and streamed Excel writes delivered by email.
- Built backend services for a Terminal Management System tracking roughly 100,000 payment devices (soundboxes, POS, MPOS) across inventory, merchant-device assignment, and firmware-version visibility.
- Implemented bulk device ingestion from Excel with BullMQ background processing and WebSocket real-time progress reporting, plus per-record validation producing a downloadable error sheet for correction and re-upload.
- Integrated Fastify services with MQTT topics driving payment-confirmation announcements on soundbox devices deployed in the field.

Intugine Technologies, Software Development Engineer I

Nov 2022 - Nov 2023 · Bengaluru

Logistics visibility platform.

- Developed features for Control Tower, a server-driven UI platform giving logistics teams network-wide shipment visibility across their supply chain.
- Owned the company web presence end to end: migrated the CMS from Ghost to WordPress with no content loss, integrated HubSpot for lead capture and marketing automation, and reduced page load times across web and mobile to improve Core Web Vitals and organic search performance.

SELECTED ENGINEERING WORK

Production OTT Platform, Video Ingest and Live Streaming

Client engagement

- Built the video ingest pipeline for a commercial VOD platform: browser-to-S3 presigned multipart upload with adaptive 5 to 100 MB chunking and a 10-worker sliding-window pool, removing the API server as a bandwidth and memory bottleneck for multi-GB film files, with resumable sessions and automated cleanup of orphaned S3 parts.
- Generated AWS MediaConvert HLS jobs from a typed rendition ladder and cut wasted transcode minutes and storage by probing source resolution in the browser and filtering out any rendition that would upscale the source; separately built AWS IVS live streaming end to end, including RTMPS channel provisioning, polling for asynchronous recording-config activation, and a webhook-driven event state machine.

OPEN SOURCE

- [@manik1848/react-date-range-picker](#) · Published React date-range picker supporting preset ranges plus absolute, relative, and "now" selection modes.
- [monorepo-bp](#) · Turborepo and pnpm reference monorepo wiring Fastify v5, Next.js 15, and Expo with end-to-end type-safe APIs.

EDUCATION

B.Tech, Textile Technology · Anna University · 2021

MERN Full-Stack Program · Masai School · 2022